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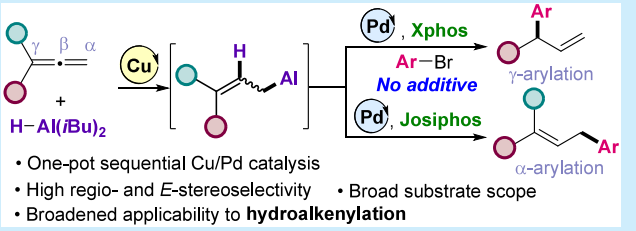
Cu/Pd-Catalyzed Ligand-Controlled Regiodivergent Hydroarylation of Allenes Using Diisobutylaluminum Hydride

Seungho Kim,‡ Kyeongmin Lee,‡ Seeun Lim, and Yunmi Lee*

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ABSTRACT: We report a ligand-controlled, regiodivergent Cu/Pd-catalyzed hydroarylation of unactivated allenes. The reaction proceeds via Cu-catalyzed hydroalumination with DIBAL-H and subsequent additive-free Pd-catalyzed cross-coupling with aryl bromides. The key feature of this work is the discovery that modulation of the palladium ligand decisively enables complete regiodivergence. Bulky monodentate Xphos ligands promote exclusive γ-selective arylation, while bidentate Josiphos-type ligands switch the pathway to afford α-selective arylation. The method features mild conditions, scalability, and broad utility for synthesizing polysubstituted aromatic compounds.



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Hydroarylation of unsaturated substrates has emerged as a powerful strategy for constructing aryl-substituted alkenes with precise control over regio- and stereochemistry.[1] These arylated alkenes represent key structural motifs in pharmaceuticals, agrochemicals, natural products, and advanced materials.[2] Accordingly, the development of efficient and selective hydroarylation methods is essential for the streamlined synthesis of complex molecular architectures. Hydroarylation has advanced significantly through transition metal catalysis, enabling high levels of regio- and stereocontrol with Pd-, Rh-, Ni-, Fe-, and Co-based catalysts, and employing diverse aryl sources such as aryl halides, arylboronic acids, and arenes.[3] However, compared with alkenes[4] and alkynes,[5] the hydroarylation of allenes[6] remains relatively underexplored. Allenes, with their unique cumulative diene structures, can provide direct access to allylic or vinylic aryl compounds, yet controlling regio- and stereoselectivity is challenging due to the formation of multiple possible isomers (Scheme 1a).[7] Conventional approaches relying on directing groups or specific substrate designs through aromatic C−H activation[8] often suffer from limited generality. Thus, the development of broadly applicable catalytic systems for regioselective hydroarylation of unactivated allenes remains a desirable goal.

Another major approach for the hydroarylation of allenes involves the use of preactivated aryl reagents for transition metal catalysis (Scheme 1b). The Ma and Oh groups independently reported Pd-catalyzed hydroarylations of allenes using arylboronic acids with catalytic acetic acid, which predominantly yielded β-arylated alkenes (Type I).[9] Yoshida et al. described a related regiocontrolled addition of arylboronic acids to allenes, demonstrating that different β-arylated regioisomers (Type I vs II) could be obtained by switching the catalyst from Pd to Pt; however, both the Pd and Pt systems still predominantly produced β-arylation products, and the substrate scope remained limited to monosubstituted allenes.[10] Shirakawa et al. disclosed the nickel-catalyzed hydroarylation of an allene with tolylboronate, which proceeded via γ-selective arylation with only moderate regioselectivity (65% γ-product and 35% α-product).[11] As another approach, Miller and Montgomery developed allene hydroarylation in which an *N*-heterocyclic carbene (NHC)−Pd catalyst promotes regioselective hydrosilylation using silanes as the hydride source to generate alkenylsilane intermediates that subsequently undergo Pd-catalyzed cross-coupling with aryl iodides in the presence of n-Bu_4NF, affording β-arylated alkenes (Type II) (Scheme 1c).[12] Despite these advances, current catalytic systems still predominantly deliver β-arylation products, and truly regiodivergent catalytic systems that enable selective access to multiple regioisomers from the same substrates remain exceedingly rare.

Herein, we report a ligand-controlled, regiodivergent hydroarylation of unactivated allenes through a dual Cu/Pd catalytic system. The reaction involves Cu-catalyzed hydroalumination of allenes with diisobutylaluminum hydride (DIBAL-H),[13] followed by Pd-catalyzed cross-coupling with aryl bromides in a single vessel (Scheme 1d). This approach delivers two distinct regioisomeric products, α- and γ-arylated alkenes (Type III and IV), from a single allene substrate, representing a rare example of true ligand-controlled regiodivergence. Although Pd-catalyzed cross-couplings of

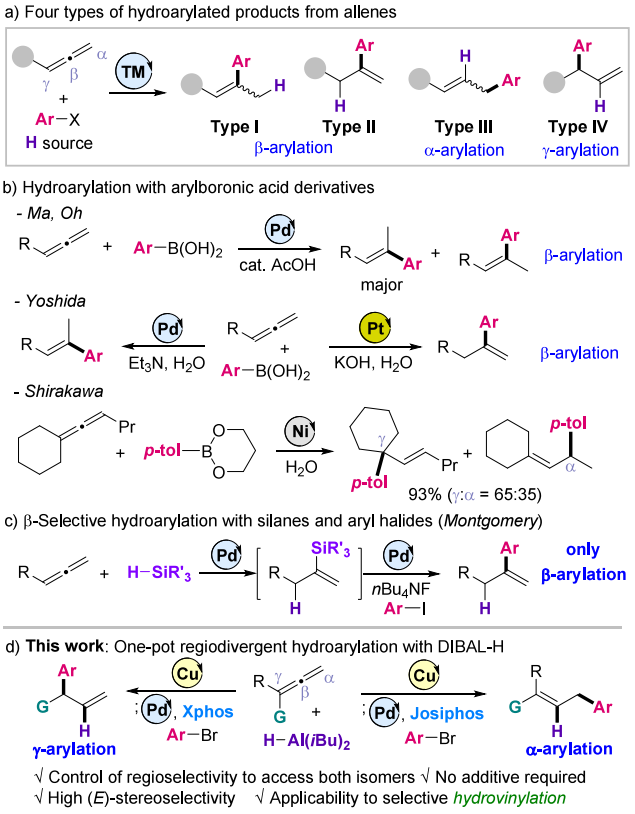
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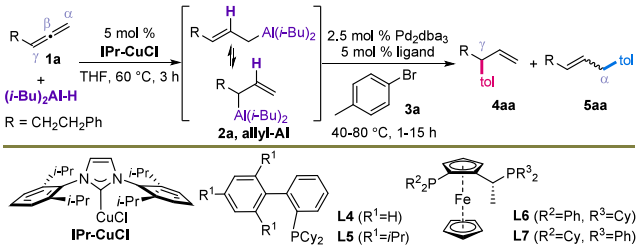
Scheme 1. Transition-Metal-Catalyzed Hydroarylation of Allenes



aryl halides with allylic organometallic reagents such as allylaluminums and allylboronates have been reported,[14] these methods typically rely on preformed metallic reagents, requiring additional synthetic steps. Moreover, allylboronate-based reactions often necessitate external additives, while the use of allylaluminum reagents has been limited due to the lack of mild and general methods for their preparation, thereby restricting substrate scope. In contrast, allenes provide a straightforward and versatile entry to allylic metal intermediates through in situ catalytic hydrometalation.

We optimized the regiodivergent hydroarylation of allene **1a** using DIBAL-H as the hydride source and 4-bromotoluene (**3a**) as the aryl coupling partner (Table 1). Allene **1a** underwent smooth hydroalumination in the presence of 5 mol % **IPr-CuCl** catalyst, forming allylaluminum intermediate **2a** with >98% conversion in situ.[13] Subsequently, we evaluated various Pd-catalyzed coupling conditions using **3a** to identify the optimal reaction parameters, focusing on the ligand choice to control the regioselectivity of the arylation step. In the absence of an additional phosphine ligand, using Pd_2dba_3 alone, no coupling product was observed (entry 1). Introduction of monodentate phosphines such as PPh_3 and PCy_3 enabled Pd-catalyzed coupling, delivering γ-arylated product **4aa** in moderate yields (32−40%) alongside α-arylated product **5aa** in 7−18% yields (entries 2 and 3). Screening of various bidentate phosphine ligands (**L1**−**L3**) resulted in a low efficiency and poor regioselectivity (entries 4−6). In contrast, the use of bulky Buchwald-type monodentate phosphine ligands (**L4** and **L5**) significantly improved both the reactivity and γ-selectivity of Pd-catalyzed coupling (entries 7 and 8). Further optimization revealed that increasing the amount of the allylaluminum intermediate and adjusting

Table 1. Optimization of Reaction Conditions[a]



entry	equiv of **2**	ligand (mol %)	temp. (°C)	time (h)	yield of **4aa**:**5aa** (%)[b]
1	1.2	no	80	12	<2:<2
2	1.2	PPh_3 (10)	80	12	40:18
3	1.2	PCy_3 (10)	80	12	32:7
4	1.2	dppBz, **L1** (5)	80	12	17:22
5	1.2	binap, **L2** (5)	80	12	20:30
6	1.2	DPEphos, **L3** (5)	80	12	29:22
7	1.2	**L4** (5)	80	12	64:5
8	1.2	**L5** (5)	80	12	67:5
9[c]	1.5	**L5** (5)	80	12	85:7
10[c]	1.5	**L5** (5)	**60**	**1**	**96:4**
11[c]	1.5	**L5** (5)	40	1	48:7
12	1.2	**L6** (5)	80	12	6:50
13	1.2	**L7** (5)	80	12	<2:72
14[c]	1.5	**L7** (5)	**80**	**15**	**<2:94**[d]

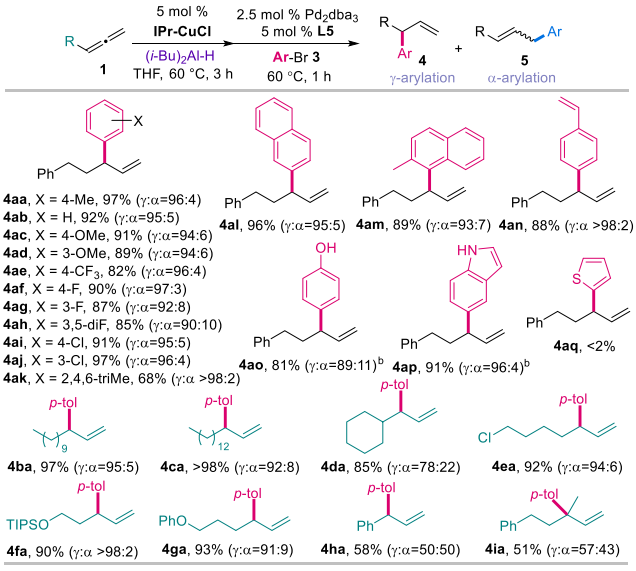
[a]Reaction conditions: allene (**1a**, 0.30 mmol), $(i\text{-Bu})_2$Al−H (0.30 mmol), **IPr-CuCl** (5 mol %), Pd_2dba_3 (2.5 mol %), ligand (5−10 mol %), tol-Br (**3a**, 0.25 mmol), THF (0.20 M). [b]Determined by ^{1}H NMR spectral analysis using 1,3,5-trimethoxybenzene as the internal standard. [c]Tol-Br **3a** was used in an amount of 0.20 mmol. [d]84:16 *E*/*Z* of **5aa**.

the reaction conditions to 60 °C for 1 h in the presence of **L5** afforded γ-arylated product **4aa** predominantly, with excellent regioselectivity (96:4 γ/α; entries 9 and 10). However, lowering the temperature to 40 °C reduced this efficiency (entry 11). Notably, shifting to the Josiphos-type bidentate ligands (**L6** and **L7**) altered the reaction outcome, favoring α-arylation (entries 12 and 13). Ultimately, the combination of ligand **L7** with the Pd_2dba_3 catalyst delivered α-arylated product **5aa** in 94% yield with an *E*/*Z* ratio of 84:16 (entry 14), thus demonstrating the capacity of ligand choice to switch the regioselectivity of the hydroarylation process. Notably, when allene **1a**, DIBAL-H, **3a**, **IPrCuCl**, and the Pd catalyst were combined in a single operation, the β-selective hydroarylation proceeded smoothly to afford a mixture of β-arylated alkenes (Type I and II, 90% yield).

With these optimal conditions, the substrate scope was examined with respect to both allenes and aryl bromides (Scheme 2). The γ-selective arylation proceeded efficiently with ligand **L5**, affording γ-arylated alkenes **4aa**−**4aj** in high yields (82−97%) and excellent regioselectivity (γ:α > 90:10) from diverse aryl bromides **3a**−**3j** bearing electron-donating or electron-withdrawing groups (e.g., MeO, CF_3, Cl, F). Sterically hindered mesityl and naphthyl bromides were also compatible, delivering **4ak**−**4am** predominantly as γ-arylated products with slightly reduced efficiency for **4ak** (68% yield). Furthermore, the reaction tolerated various functional groups on aryl bromides, such as alkenes and alcohols, affording **4an** and **4ao** in good yields. The method was also applicable to heteroaryl bromides: indolyl bromide underwent smooth

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Scheme 2. Scope for γ-Selective Hydroarylation[a]



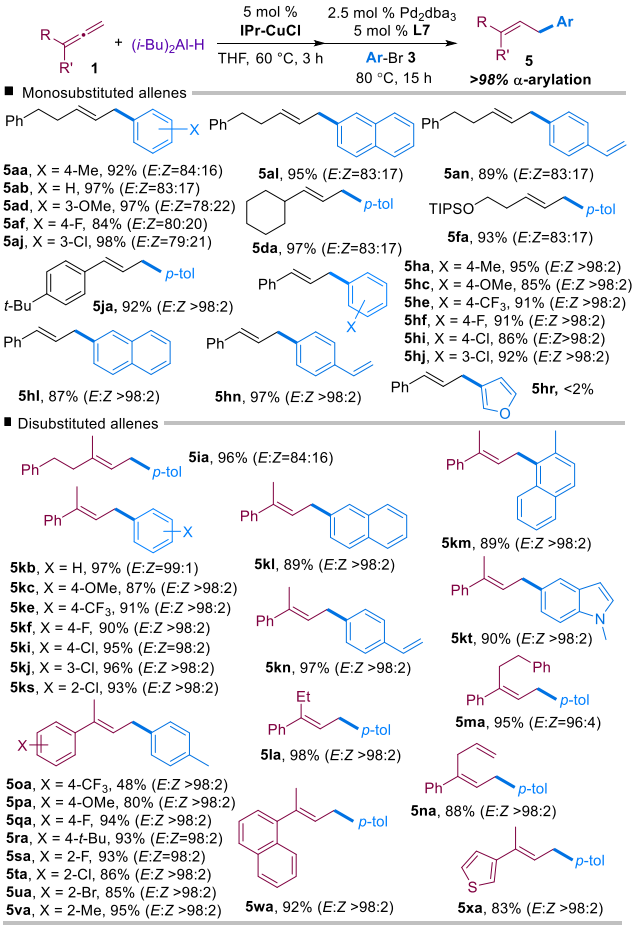
[a]Reaction conditions: allene (**1**, 0.30 mmol), (*i*-Bu)$_2$Al−H (0.30 mmol), Ar−Br (**3**, 0.20 mmol), THF (0.20 M). Yields of the isolated products were obtained. [b]Allene and DIBAL-H were used as 2.5 equiv each. Under the standard conditions, the yields of **4ao** and **4ap** were 25% and 56%, respectively.

coupling to give **4ap** in 91% yield, whereas thienyl bromide failed to afford **4aq**, likely due to the incompatibility of sulfur-containing heterocycles under the reaction conditions.[15] Alkyl-substituted allenes **1b**−**1d** bearing long-chain nonyl, tridecyl, or bulky cyclohexyl groups underwent γ-arylation to afford **4ba**−**4da** in high yields, though **1d** gave moderate regioselectivity (γ:α = 78:22). Functionalized allenes **1e**−**1g** bearing chloro, silyl ether, or phenyl ether groups were well tolerated, providing **4ea**−**4ga** with consistently high γ-selectivity. In contrast, phenylallene **1h** and disubstituted allene **1i** exhibited diminished selectivity, giving nearly statistical mixtures of regioisomers (γ:α = 50:50 and 57:43, respectively).

Next, the scope of the α-selective arylation was examined (Scheme 3). All transformations furnished the α-arylated products with >98% regioselectivity and good-to-excellent *E*-selectivity. Phenethyl allene **1a** in the presence of ligand **L7** reacted smoothly with diverse aryl bromides bearing Me, MeO, F, Cl, alkene, or naphthyl groups to afford **5aa**−**5an** in high yields (84−98%, *E*:*Z* > 78:22). Cyclohexyl (**1d**) and silyl ether-substituted (**1f**) allenes also underwent α-arylation efficiently, providing **5da** and **5fa**. Notably, aryl allene **1j** gave **5ja** in 92% yield with >98% *E*-selectivity. Furthermore, phenylallene **1h** reacted with various aryl bromides (Me, MeO, CF_3, F, Cl, naphthyl, or alkene groups) to furnish α-arylated products **5ha**−**5hn** in 85−99% yields, whereas coupling with furyl bromide was ineffective (<2% yield of **5hr**), instead producing dimerization products derived from the allylaluminum intermediate **2h** (Scheme S1).

In addition, α-selective hydroarylation of disubstituted allenes was examined, predominantly yielding *E*-trisubstituted alkenes. Dialkyl allene **1i** underwent exclusive α-arylation to give **5ia** in 96% yield (*E*:*Z* = 84:16). Disubstituted allenes bearing aryl groups consistently afforded single regioisomers with excellent *E*-selectivity (>98%). This transformation provides an efficient route to *E*-trisubstituted alkenes,

Scheme 3. Scope for α-Selective Hydroarylation[a]



[a]Reaction conditions: allene **1** (0.30 mmol), (*i*-Bu)$_2$Al−H (0.30 mmol), Ar−Br **3** (0.20 mmol), THF (0.20 M). Yields of the isolated products were obtained.

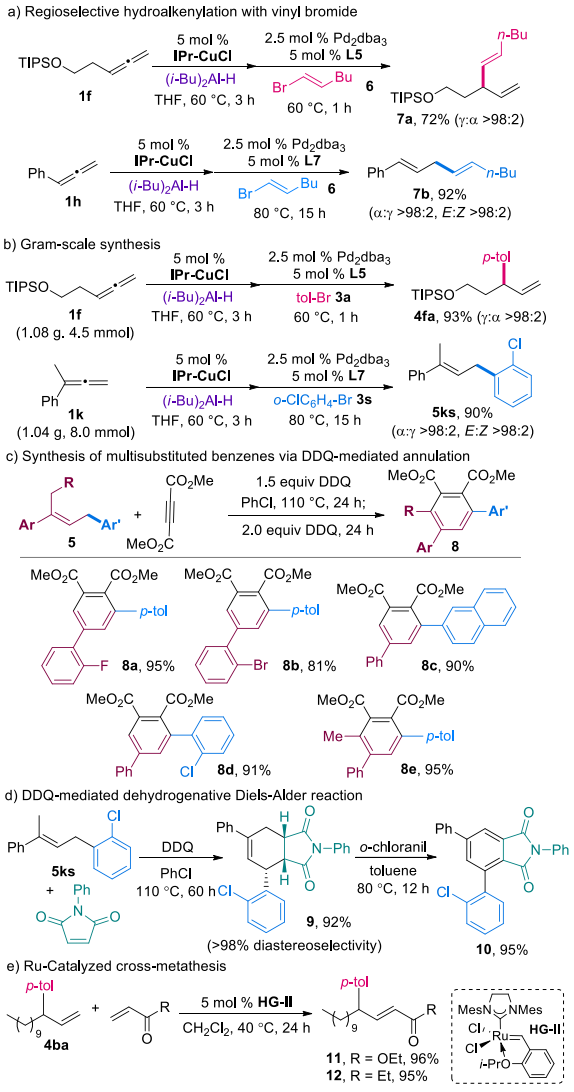
challenging motifs in pharmaceuticals, natural products, and functional materials.[16] The α-arylation of phenyl, methyl-substituted allene **1k** with various aryl- and naphthyl bromides bearing MeO, CF_3, F, Cl and alkene substituents afforded α-arylated *E*-alkenes **5kb**−**5kn** in 87−97% yields. Indolyl bromide was also tolerated to furnish **5kt** in 90% yield. Disubstituted allenes **1l**−**1n** bearing phenyl and alkyl groups such as ethyl, phenethyl, or allyl groups provided **5la**−**5na** in 88−98% yields. Similarly, aryl-substituted allenes **1p**−**1v** with MeO, *t*-Bu, Me, F, Cl, or Br groups yielded **5pa**−**5va** in 80−95% yields. Interestingly, CF_3-phenyl allene **1o** gave **5oa** in moderate yield (48%) due to unexpected dimerization side reactions, whereas naphthyl and thienyl allenes **1w** and **1x** afforded **5wa** (92%) and **5xa** (83%), respectively.

We also investigated the regiodivergent hydroalkenylation of allenes using vinyl bromides **6** (Scheme 4a). While the alkenylation of **1f** with vinyl bromide **6** in the presence of **L5** provided γ-alkenylated product **7a** in 72% yield, the use of the **L7** ligand enabled α-alkenylation of **1h**, affording *E*-selective diene product **7b** in 92% yield.

The practical utility and scalability of this method were further demonstrated by the successful gram-scale syntheses of **4fa** and **5ks** (Scheme 4b). The γ-arylation of allene **1f** (1.08 g) proceeded smoothly to deliver the γ-arylated alkene **4fa** in 93% yield. Likewise, the α-arylation of allene **1k** (1.04 g) afforded

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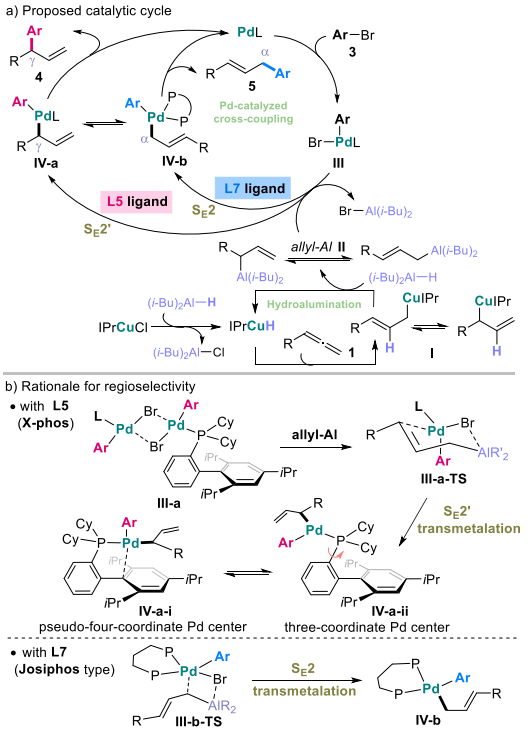
Scheme 4. Synthetic Applications



the *E*-trisubstituted alkene **5ks** in 90% yield. To further highlight the applicability of our method, a one-pot transformation of trisubstituted alkene **5sa** into a polysubstituted aromatic compound was investigated. Treatment of **5sa** with 2,3-dichloro-5,6-dicyanobenzoquinone (DDQ) and dimethyl acetylenedicarboxylate under heating conditions induced in situ oxidation to a diene intermediate, which subsequently underwent a Diels−Alder reaction followed by oxidation to afford tetrasubstituted arene **8a** in 95% yield (Scheme 4c). Various trisubstituted alkenes were successfully transformed into the corresponding arenes **8b**−**8d** in 81−91% yield, along with the formation of the pentasubstituted arene **8e** in 95% yield. Additionally, the tandem oxidative annulation of **5ks** with *N*-phenylmaleimide in the presence of DDQ proceeded via Diels−Alder cycloaddition, furnishing cycloadduct **9** in 92% yield (Scheme 4d).[17] The subsequent oxidation of **9** with *o*-chloranil provided the aromatic compound **10** in 95% yield. Moreover, the synthetic versatility of the γ-arylated products was further demonstrated by Ru-catalyzed metathesis of alkene **4ba** with ethyl acrylate or ethyl vinyl ketone in the presence of the Hoveyda−Grubbs catalyst, which afforded the corresponding *trans*-disubstituted alkenes **11** and **12** in excellent yields (Scheme 4e).

The proposed mechanism (Scheme 5a) involves initial Cu-catalyzed hydroalumination of allene **1** with DIBAL-H,

Scheme 5. Plausible Mechanism



generating allylaluminum intermediate **II**[13] via regioselective insertion of in situ-formed **IPrCuH**.[18] Subsequent Pd-catalyzed transmetalation between **II** and the aryl-Pd(II) complex **III** delivers σ-allylpalladium intermediates **IV** through either the S_E2 or S_E2' pathway, depending on the ligand.[19] Reductive elimination then furnishes the α- or γ-arylated product under ligand conotrol.[14] Although the reaction likely proceeds via direct reductive elimination, interconversion between σ- and π-allylpalladium intermediates cannot be completely ruled out.[20] With bulky, electron-rich Xphos **L5**, the conformation of aryl-Pd(II) complex **III-a**, supported by an X-ray crystal structure,[21] provides steric accessibility for the allylaluminum to approach the Pd center and facilitates S_E2' transmetalation through a six-membered transition state **III-a-TS** (Scheme 5b). The resulting three-coordinate Pd(II) complex **VI-a-ii** rearranges to a pseudo-four-coordinate species **VI-a-i**[22] that undergoes rapid reductive elimination to afford the γ-arylated product **4**. In contrast, with the chelating ligand **L7**, the S_E2 pathway (**III-b-TS**) is favored, forming σ-allyl-Pd(II) complex **IV-b** and yielding α-arylated products. The rigid bite angle and constrained coordination geometry imposed by the **L7** ligand[23] likely stabilize the S_E2 transition state and suppress σ−π−σ isomerization, resulting in high α-selectivity. Thus, ligand-dependent regioselectivity arises from the preferred transmetalation pathway (S_E2 vs S_E2'), dictated by the coordination environment around palladium.

In summary, we have developed a Cu/Pd dual-catalyzed, ligand-controlled regiodivergent hydroarylation of unactivated allenes using DIBAL-H and aryl bromides. This approach enables the selective synthesis of otherwise inaccessible α- and γ-arylated alkenes from a common allene substrate. The regioselectivity is precisely dictated by the phosphine ligand in

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the Pd-catalyzed step, allowing mild, broad-scope reactions with excellent regio- and stereocontrol. The same strategy is also effective for hydroalkenylation, furnishing 1,4-dienes with high selectivity, and the synthetic versatility of the products was showcased via Diels–Alder/oxidative aromatization sequences. This study establishes a new ligand-directed platform for achieving regiodivergent C–C bond formation from allenes beyond the reach of conventional hydroarylation chemistry.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c04396.

Experimental procedures, characterization of products, and spectral data (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Yunmi Lee − *Department of Chemistry, Kwangwoon University, Seoul 01897, Republic of Korea;* orcid.org/0000-0003-1315-4001; Email: ymlee@kw.ac.kr

Authors

Seungho Kim − *Department of Chemistry, Kwangwoon University, Seoul 01897, Republic of Korea*

Kyeongmin Lee − *Department of Chemistry, Kwangwoon University, Seoul 01897, Republic of Korea*

Seeun Lim − *Department of Chemistry, Kwangwoon University, Seoul 01897, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.5c04396

Author Contributions

‡S.K. and K.L. contributed equally.

Notes

The authors declare no competing financial interest.

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